Abu Henaf Rashid Ahmad Mufti

Applied AI Engineer & Research Enthusiast

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CAREER OBJECTIVE

Applied AI engineer and research enthusiast working across machine learning, ensemble methods, and explainable AI. Builds end-to-end pipelines that turn complex, multi-modal data into structured, interpretable models, and applies the same rigor to production-facing LLM and RAG systems. Focused on healthcare, environmental, and biological problem domains, with a growing interest in scalable, real-world AI deployment.

EDUCATION

United International University — BSc in Computer Science and Engineering *Final Year Student*
Academia School, Mirpur, Dhaka — A Level & O Level, Edexcel Curriculum (Edexcel Award, Daily Star Award)

SKILLS

Machine Learning & AI: Ensemble Learning, Neural Networks (1D CNN), Model Explainability, Feature Engineering

Applied AI / LLM Engineering: RAG pipelines, LangGraph agent orchestration, multi-model prompt chaining, vector retrieval (ChromaDB), multi-modal (vision + voice) pipelines

Data Science: Statistical Analysis, Data Mining, Predictive Modelling, Multi-modal Data Processing, Image Data Analysis

Programming & Frameworks: Python, PyTorch, C++, React, Node.js, FastAPI, Streamlit

Database & Query: SQL, Database Handling, Data Pipeline Design

Soft Skills: Fluent Spoken and Written English, Adaptive and Observant, Methodological Idea Generation, Complex Problem Simplification

PUBLICATIONS

An Explainable and Scalable Framework for Compost Maturity Prediction (ACM Digital Library) *2024*
Ensemble Learning and Conversational AI

– Developed a scalable framework integrating Ensemble Learning with Conversational AI to produce interpretable, explainable predictions for agricultural waste management.

RESEARCH

A Multimodal Risk Assessment and Monitoring Framework for Assisted Elderly Living *2026 – Present*
Final Year Design Project (FYDP)

– Designed a multimodal pipeline for continuous observation of physiological and behavioural indicators to detect potential health risks and emergencies in assisted living environments.
– Platform provides timely alerts and health insights for caregivers through signal-based ensemble classification across heterogeneous sensor data.

DNA Protein-Site Binding Evaluation *2025 (Mid) – Finalised*
Machine Learning Project

– Applied ML architecture to predict protein-site binding from biological sensorgram data, simplifying high-dimensional biological signals into interpretable tabular formats.

Pneumonia Lung Disease Classification *2025 (Mid) – Finalised*
Digital Image Processing Project

– Developed an image-specific data retrieval and segmentation pipeline for GPU-scarce environments, converting medical image data into structured formats for diagnostic classification.

AQI Prediction and Generalisation *2026 – Present*
Peer Collaboration

– Applied Ensemble Learning on structured environmental datasets to predict air quality indices, prioritising model generalisation and public interpretability of outputs.

PROJECTS

Unified-MediChat — GitHub
Multi-mode FastAPI health assistant with consultant, therapy, and vision+voice pipelines, backed by RAG, Groq, and ElevenLabs.

Data-Seeker — GitHub
Natural language to SQL workspace with a validated query pipeline and a second LLM pass that writes an analyst-style brief.

Research_Assist — GitHub
AI-powered research assistant that searches, organizes, and summarizes academic papers to streamline literature reviews.

Context_ChatBot — GitHub
Streamlit RAG assistant that answers questions from uploaded PDFs and falls back to live web search when needed.

Dermatologist_LLMCHAT — GitHub
LLM-powered chatbot for answering dermatology-related questions and providing educational guidance on skin conditions.

Web-ScrapChat — GitHub
Full-stack search-chat app with a Next.js frontend, FastAPI backend, and LangGraph-driven real-time streaming search.

PROJECT ROLES

- Methodology — Designing end-to-end data pipelines for efficient structured processing and model training.
- Architecture — Integrating ML and neural network models within software and data infrastructure.
- Analysis — Transforming high-dimensional data into interpretable formats for scientific reporting and insight generation.